

Tanvitha Doradla

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Professional Summary

Data Science and Machine Learning Engineer with over 2 years of experience in developing predictive analytics, time series forecasting, and deep learning models. Proven ability to design end-to-end data pipelines, deploy ML models using Flask/Django, and analyze large datasets using Python, SQL, and cloud platforms like AWS and Azure. Demonstrated expertise in NLP (LLM fine-tuning, RLHF), image classification, and business intelligence reporting with Tableau and Power BI. Strong foundation in statistical modeling, big data tools (Spark, Databricks), and data engineering best practices.

Technical Skills

Machine Learning & Deep Learning: KNN, Decision Trees, SVM, Random Forest, Gradient Boosting, ARIMA, PCA, LDA, ANN, CNN, RNN, LSTM

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

NLP & LLMs: LLMs (e.g., LLaMA), Prompt Engineering, RLHF, Corpus Labeling, Sentiment Analysis

Data Visualization: Tableau, Power BI, Matplotlib, Seaborn

Programming Languages: Python (Primary), R, SQL, Java (Core Java, OOP)

Databases: MySQL, PostgreSQL, Oracle, MongoDB, SQLite

Cloud & Big Data Tools: AWS (EC2, S3, Lambda), Azure (App Services, VMs), Hadoop, Spark, Databricks, AWS Glue

ETL & Data Pipelines: Python/SQL-based ETL workflows, API/JSON/XML integrations, parallel/distributed data processing

Development Tools: GitHub, PyCharm, Flask, Django

Work Experience

Data Science Intern

SMG Infotech, India

Apr 2021 – Jul 2022

- Built multiple supervised and unsupervised machine learning models (Random Forest, SVM, KNN) to predict customer churn, improving prediction accuracy by 18%.
- Developed a price forecasting engine using ARIMA and LSTM models, reducing revenue prediction errors by over 22%.
- Implemented feature engineering and normalization techniques using Pandas and NumPy on large healthcare datasets (~5M records).
- Designed dashboards using Power BI and Tableau to visualize key metrics for operational and clinical decision-making.
- Conducted exploratory data analysis (EDA) and outlier detection for various domains, improving data quality and integrity by 30%.
- Deployed ML models into production using Flask and integrated with web applications for client-facing tools.

- Automated ETL pipelines using Python and SQL, streamlining data ingestion from APIs and relational databases.
- Collaborated cross-functionally with software engineers and stakeholders to align analytics efforts with business goals.

Data Science Intern (Unpaid)

ADEPT COMPUTER CONSULTANTS INC., USA

Aug 2023 – Present

- Fine-tuned pre-trained LLMs (e.g., LLaMA) for text classification and lead intent detection tasks using RLHF and prompt engineering.
- Built NLP pipelines for sentiment analysis and entity recognition on marketing campaign data using Python, spaCy, and Hugging Face Transformers.
- Designed and trained deep learning models (CNN, RNN) for image classification tasks related to A/B ad performance testing.
- Optimized data preprocessing scripts using parallel processing in Pandas and Dask, reducing pipeline runtime by 45%.
- Analyzed SQL-based marketing data to identify conversion trends, contributing to a 12% increase in customer acquisition.
- Created interactive reports using Tableau for C-level stakeholders to track real-time campaign effectiveness.
- Implemented model versioning and reproducibility practices using GitHub, MLflow, and PyTorch Lightning.
- Assisted in integrating ML models with internal APIs using Flask, ensuring seamless handoff to engineering teams.

Education

Master of Science in Data Science

University of Texas at Arlington, USA, Aug 2022 – May 2024

- **Relevant Coursework:** Machine Learning, Deep Learning, Big Data Analytics, Statistical Modeling, Cloud Computing
- **Projects:**
 - **LLM Fine-Tuning & Deployment:** Fine-tuned transformer-based LLMs for intent classification and document summarization.
 - **Time Series Forecasting for Retail Demand:** Built ARIMA and LSTM models using sales data to improve stock predictions by 20%.
 - **NLP Chatbot with Prompt Engineering:** Developed a custom FAQ bot for academic use using prompt-tuned LLMs.

Bachelor of Technology in Computer Science Engineering

Vellore Institute of Technology, Amaravati, Andhra Pradesh, India, May 2022

- **Relevant Coursework:** Data Structures, Algorithms, Database Systems, Software Engineering